

# Notes on Lattices

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## Abstract

Notes taken while studying lattice-based cryptography.

Usually while reading books and papers I take handwritten notes in paper notebooks, this document contains some of them re-written to *LaTeX*.

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## 1 Lattices

### 1.1 Hard problems

**Definition 1.1** (SVP (Shortest Vector Problem)). find a shortest non-zero vector  $x \in \mathcal{L}$ , ie. that minimizes the Euclidean norm  $\|x\|$ , that is  $\lambda_1(\mathcal{L}) = \|x\|$ .

**Definition 1.2** (CVP (Closest Vector Problem)). given a vector  $y \in \mathbb{R}^n$  with  $y \notin \mathcal{L}$ , find a nonzero vector  $x \in \mathcal{L}$  that is closest to  $y$ , ie. that minimizes the Euclidean norm  $\|y - x\|$ .

**Definition 1.3** (NTRU key recovery problem). Given  $h(x)$ , find ternary polynomials  $f, g \in \mathcal{T}$  satisfying

$$f(x) \cdot h(x) = g(x) \pmod{q} \in R_q$$

Where  $R_q = \mathbb{Z}_q[X]/(X^n - 1)$ .

**Definition 1.4** (SIS (Short Integer Solution)). Given  $q \in \mathbb{Z}$  and matrix  $A \in \mathbb{Z}_q^{m \times n}$ , find a non-zero short  $z \in \mathbb{Z}^n$  s.th.

$$A \cdot z = 0 \pmod{q}$$

"short":  $z \in \{B, B\}^n$  for some bound  $B$ , often  $z \in \{-1, 0, 1\}^n$ .

The set of all solutions to the SIS problem forms the (q-ary) lattice

$$\Lambda_q^\perp(A) = \{z \in \mathbb{Z}^n \mid Az = 0 \in \mathbb{Z}_q^m\} \supseteq \mathbb{Z}_q^n$$

where the SIS problem becomes the SVP for  $\Lambda_q^\perp(A)$

- LLL algorithm is used to solve SVP and CVP.
- Both SVP and CVP become computationally difficult (NP hard) as the lattice dimension  $n$  grows.

**Definition 1.5** (Ring SIS (Ring Short Integer Solution)). given a uniformly random matrix  $A \in^R R_q^n$ , with elements  $a_i \in R_q$ , find a non-zero integer vector  $z \in R^n$  of norm  $\|z\| \leq \beta$  such that

$$Az = \sum_i a_i \cdot z_i = 0 \in R_q$$

**Definition 1.6** (MSIS (Module Short Integer Solution)). given a uniformly random matrix  $A \in^R R_q^{m \times n}$ , with columns  $a_i \in R_q^m$ , find a non-zero integer vector  $z \in R^n$  of norm  $\|z\| \leq \beta$  such that

$$Az = 0 \in R_q^m$$

$$\text{ie. } \sum_i a_{ij} \cdot z_j = 0 \in R_q, \text{ for } i = 1, \dots, m$$

Notice that Module-SIS is a Ring-SIS with  $rank = 1$  ( $m = 1$ ).

The next form is useful to compress the size of a SIS matrix by  $n$  columns:

**Definition 1.7** (Hermite Normal Form (HNF)). Assume the leftmost  $n$  columns  $A_1 \in \mathbb{Z}_q^{n \times n}$  of  $A = [A_1 | A_2] \in \mathbb{Z}_q^{n \times m}$  form an invertible matrix.

Then replace  $A$  by

$$A_1^{-1} \cdot A = [I_n \mid \bar{A} = A_1^{-1} \cdot A_2]$$

and treat the  $I_n$  submatrix as implicit.

Then,  $A$  and  $[I_n \mid \bar{A}]$  have exactly the same set of short SIS solutions.

**Definition 1.8** (LWE distribution).  $s \in \mathbb{Z}_q^n$  (secret),  $a \in \mathbb{Z}_q^n$ ,  $e \in \Xi$ , output

$$(a, b = \langle s, a \rangle + e \pmod{q}) \in (\mathbb{Z}_q^n \times \mathbb{Z}_q)$$

**Definition 1.9** (Search-LWE). given  $m$  independent samples  $(a_i, b_i) \in \mathbb{Z}_q^n \times \mathbb{Z}_q$ , for a uniform  $s \in \mathbb{Z}_q^n$  (fixed for all samples), find  $s$ .

**Definition 1.10** (Decision-LWE). given  $m$  independent samples  $(a_i, b_i) \in \mathbb{Z}_q^n \times \mathbb{Z}_q$ , distinguish which one is not random and it is an actual instance.

## 1.2 Dual lattices

From previous section, we have LWE-Lattice and SIS-Lattice:

Search-LWE can be seen as a Bounded-Distance Decoding (BDD) problem on  $q$ -ary  $m$ -dimensional integer lattices:  
for LWE samples, vector  $b$  is relatively close to exactly one vector in the LWE lattice:

**Definition 1.11** (LWE Lattice). A LWE-Lattice  $\mathcal{L}$  is defined

$$\begin{aligned}\mathcal{L} &= \{A^T s \mid s \in \mathbb{Z}_q^n\} + q\mathbb{Z}^m \\ &= \{A^T s + qz \mid s \in \mathbb{Z}_q^n, z \in \mathbb{Z}^m\}\end{aligned}$$

SIS can be seen as a SVP on  $q$ -ary  $m$ -dimensional lattices:

**Definition 1.12** (SIS-Lattice). A SIS-Lattice  $\mathcal{L}$  is defined

$$\mathcal{L}^\perp = \{z \in \mathbb{Z}^m \mid Az = 0 \in \mathbb{Z}_q^n\} \supseteq q\mathbb{Z}^m$$

**Definition 1.13** (Dual of a lattice). For a lattice  $\mathcal{L}$ , its dual is

$$\mathcal{L}^* = \{y \in \mathbb{R}^m \mid \langle y, x \rangle \in \mathbb{Z} \ \forall x \in \mathcal{L}\}$$

The dual lattice consists of all vectors that measure lattice points with integer-valued linear measurements.

**Remark 1.14.** For a lattice basis  $B = (b_1, \dots, b_n)$ , the dual basis  $B^* = (b_1^*, \dots, b_n^*)$  is the unique basis satisfying

$$\langle b_i^*, b_j \rangle = \delta_{ij} = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j \end{cases}$$

where  $\delta_{ij}$  is the Kronecker delta.

**Lemma 1.15.**

$$\mathcal{L} = q\mathcal{L}^{\perp*} \text{ ie. } \frac{1}{q}\mathcal{L} = \mathcal{L}^{\perp*}$$

*Proof.* i. Take  $y \in \frac{1}{q}\mathcal{L}$ , then by 1.11,  $\exists s \in \mathbb{Z}_q^n, z \in \mathbb{Z}^m$  such that

$$y = \frac{1}{q}(A^T s + qz) = \frac{1}{q}A^T s + z$$

Let  $x \in \mathcal{L}^\perp$ . By definition (1.12)  $Ax = 0 \pmod{q}$ , therefore  $\exists k \in \mathbb{Z}^n$  such that  $Ax = qk$ .

Compute

$$\begin{aligned}
\langle y, x \rangle &= \left\langle \frac{1}{q} A^T s + z, x \right\rangle \\
&= \frac{1}{q} \langle A^T s, x \rangle + \langle z, x \rangle \\
&= \frac{1}{q} \langle s, Ax \rangle + \langle z, x \rangle \\
&= \frac{1}{q} \langle s, qk \rangle + \langle z, x \rangle \\
&= \langle s, k \rangle + \langle z, x \rangle
\end{aligned}$$

and  $\langle s, k \rangle + \langle z, x \rangle \in \mathbb{Z}$ , since each of  $s, k, z, x \in \mathbb{Z}$  separately.

Thus  $\langle y, x \rangle \in \mathbb{Z}$ , hence by 1.13,  $y \in \mathcal{L}^{\perp*}$ .

We started from  $y \in \frac{1}{q}\mathcal{L}$ , so, now we have

$$y \in \mathcal{L}^{\perp*} \quad \forall y \in \frac{1}{q}\mathcal{L}$$

therefore

$$\mathcal{L}^{\perp*} \supseteq \frac{1}{q}\mathcal{L}$$

ii. Now we prove the reverse inclusion. Take  $y \in \mathcal{L}^{\perp*}$ .

Observe that  $q\mathbb{Z}^m \subseteq \mathcal{L}^{\perp}$ , because

$$\forall z \in \mathbb{Z}^m, \quad A(qz) = qAz = q0 = 0 \pmod{q}$$

Since  $y \in \mathcal{L}^{\perp*}$ , we must have  $\langle y, qz \rangle \in \mathbb{Z} \quad \forall z \in \mathbb{Z}^m$ .

Let  $e_i \in \mathbb{Z}^m$  be the  $i$ -th standard basis vector, ie.  $e_i = (0, \dots, 0, 1, 0, \dots, 0)$  (where one is at the  $i$ -th position).

We know that  $\langle y, qz \rangle \in \mathbb{Z} \quad \forall z \in \mathbb{Z}^m$  (including  $z = e_i$ ), thus, for  $qe_i = (0, \dots, 0, q, 0, \dots, 0)$ ,

$$\langle y, qe_i \rangle = q\langle y, e_i \rangle = q \cdot y_i$$

since  $\langle y, e_i \rangle = y_i$  (extracts the  $i$ -th position from  $y$ ).

So, since  $\langle y, qz \rangle \in \mathbb{Z} \quad \forall z \in \mathbb{Z}^m$ , and  $\langle y, qe_i \rangle = qy_i$ , then  $qy_i \in \mathbb{Z}$

This argument works  $\forall i \in [m]$ , that is,  $qy_1, \dots, qy_m \in \mathbb{Z}$ . Therefore

$$qy \in \mathbb{Z}^m \quad \forall y \in \mathcal{L}^{\perp*}$$

For  $\bar{x} \in \ker(A) \subseteq \mathbb{Z}_q^m$ , since  $A\bar{x} = 0$ , then  $Ax = 0 \pmod{q}$ , thus  $x \in \mathcal{L}^{\perp}$  (by its definition 1.12).

Since  $y \in \mathcal{L}^{\perp*}$ , by the definition of  $\mathcal{L}^{\perp*}$  (1.13),  $\langle y, x \rangle \in \mathbb{Z}$ . Multiply it by  $q$ ,

$$q\langle y, x \rangle = \langle qy, x \rangle \in q\mathbb{Z}$$

hence  $\langle qy, x \rangle = 0 \pmod{q}$ , thus  $qy \pmod{q} \in \ker(A)^{\perp}$ .

We'll now prove the following identity, valid for every integer  $q \geq 2$ , which we will use later:

$$(\ker A)^\perp = \text{im}(A^T)$$

where  $A : \mathbb{Z}_q^m \rightarrow \mathbb{Z}_q^n$ .

Let  $y \in \text{im}(A^T)$ . Then  $\exists s \in \mathbb{Z}_q^n$  such that  $y = A^T s$ .

Then,  $\forall x \in \ker(A)$ , by definition of the kernel,  $Ax = 0$ . Want to prove that  $y$  is orthogonal to every  $x \in \ker(A)$ :

$$\langle y, x \rangle = \langle A^T s, x \rangle = \langle s, Ax \rangle = 0$$

thus,  $\langle y, x \rangle = 0 \forall x \in \ker(A)$ , hence  $y$  is orthogonal to every  $x$ , that is,  $y \in (\ker(A))^\perp$ .

Therefore,

$$\text{im}(A^T) \subseteq (\ker(A))^\perp$$

For the reverse inclusion, it suffices to show that both sets have the same number of elements:

By the First Isomorphism Theorem,

$$|\text{im}(A)| = \frac{|\mathbb{Z}_q^m|}{|\ker(A)|} = \frac{q^m}{|\ker(A)|}$$

Notice that for every subgroup  $K \subseteq \mathbb{Z}_q^m$ , the dot product on  $\mathbb{Z}_q^m$  is nondegenerate, therefore

$$|K^\perp| = \frac{q^m}{|K|}$$

so, applied to  $\ker(A)$ ,

$$|\ker(A)^\perp| = \frac{q^m}{|\ker(A)|} = |\text{im}(A)|$$

which, by transposition properties,  $|\text{im}(A)| = |\text{im}(A^T)|$ , thus

$$|\ker(A)^\perp| = |\text{im}(A^T)|$$

hence,

$$(\ker(A)^\perp) = \text{im}(A^T)$$

Continuing from  $qy \pmod q \in (\ker(A))^\perp$ , we've just seen that  $(\ker(A))^\perp = \text{im}(A^T)$ , thus

$$\exists s \in \mathbb{Z}_q^n \text{ s.t. } qy = A^T s \pmod q$$

Therefore, for some

$$z \in \mathbb{Z}^m, \quad qy = A^T s + qz$$

which means that  $qy \in \mathcal{L}$ , by definition of  $\mathcal{L}$  (1.11). Divide that by  $q$ , obtaining

$$y \in \frac{1}{q}\mathcal{L}$$

Thus,  $\forall y \in \mathcal{L}^{\perp*}$ , we have  $y \in \frac{1}{q}\mathcal{L}$ .

Hence  $\mathcal{L}^{\perp*} \subseteq \frac{1}{q}\mathcal{L}$ .

Putting together (i).  $\mathcal{L}^{\perp*} \supseteq \frac{1}{q}\mathcal{L}$ , and (ii).  $\mathcal{L}^{\perp*} \subseteq \frac{1}{q}\mathcal{L}$ , we have

$$\mathcal{L}^{\perp*} = \frac{1}{q}\mathcal{L}$$

□

## References

- [1] Chris Peikert. A decade of lattice cryptography, 2016.
- [2] Matan Prasma. Lattices, an elementary and rigorous account, 2024.
- [3] Daniele Micciancio. Introduction to Lattices, 2024.
- [4] Daniele Micciancio and Oded Regev. Lattice-based Cryptography, 2008.
- [5] Alfred Menezes. A Gentle Introduction to Lattice-Based Cryptography, 2026.
- [6] Cynthia Dwork. Lattices and Their Application to Cryptography, 1998.